

OCEAN-SHIELD — MARITIME ENVIRONMENTAL ANALYTICS

SIH26143 PROTOTYPE • ANALYST-REVIEW CASE SUMMARY

AUTOMATED SCREENING OUTPUT — NOT A STATUTORY NOTICE OR FINDING OF LIABILITY
Requires calibrated imagery, source provenance, qualified analyst review, and competent-authority action.

Dossier Ref:	ICG-MEP-DOSSIER-20260922-GULF	Date of Issue:	22 Sep 2026, 09:14 UTC
Maritime Sector:	Gujarat Offshore / Northern Arabian Sea	Review status:	SCREENING LEAD — REVIEW REQUIRED
Target Vessel:	MT NEPTUNE GLORY	IMO / MMSI:	9384722 / 636019482
Flag State:	Liberia	Attribution rank:	40.0/100 lead-priority score

1. SATELLITE RADAR REMOTE SENSING (SAR) SCREENING

Sensor Platform:	Sentinel-1A C-Band SAR	Acquisition Timestamp:	2026-09-04 13:10:00 UTC
Swath Mode & Pol:	Interferometric Wide (IW) Swath / VV (Co-polarization optimal for sea surface roughness)	Ground Resolution:	10.0 meters/pixel
Observed Slick Centroid:	22.5845° N, 69.1861° E	Total Spill Surface Area:	0.62 km²
Mass / thickness:	Not estimated from this SAR scene	Geometry screen:	90.0/100 uncalibrated geometry screen
ADIOS Weathering State:	Not assessed: oil profile and mass are required	Evaporative Loss / Mousse:	N/A% Evaporated / N/A% Water Uptake

2. CONDITIONAL HYDRODYNAMIC TRANSPORT SCENARIO

Using an advection-diffusion Lagrangian numerical formulation with the supplied met-ocean field ($\Delta u=0.28$ m/s, $\Delta v=0.18$ m/s) and Ekman surface windage (3.2% Stokes drift factor with 15° Coriolis deflection), the slick plume trajectory was hindcasted backwards in time. The model propagates the supplied **age hypothesis** backward through the validated input field. It does not infer a release time or establish a source location; uncertainty must be quantified against authoritative current, wind, and imagery inputs before operational use:

Conditional backtracked point:	22.38500° N, 69.11500° E	Assumed slick age:	Not supplied
Conditional time offset:	T – 10.5h (not an inferred discharge time)	Total Hydrodynamic Drift:	14.80 km displacement

3. AIS CORRIDOR SCREENING & NAVIGATION CONTEXT

Analysis of **6 vessels** operating in the sector filtered down to **3 corridor candidates**. Vessel **MT NEPTUNE GLORY** is the highest-priority review lead based solely on spatio-temporal proximity. This is not an allegation or responsibility finding:

Evaluation Metric	Observed Telemetry Value	Review Context	Screen Value
Closest Point of Approach (CPA)	7.22 NM (13.37 km)	Modelled proximity to candidate origin	0.0/100
Temporal Coincidence	$\Delta t = 0.00$ hours from origin	Modelled temporal alignment	100.0/100
Speed Over Ground (SOG)	Drop: 14.7 → 5.1 kts	Large speed change (14.7 → 5.1 kts); requires navigation-context review	70.0/100
Vessel class	Crude / Product Tanker (DWT: 158000)	Descriptive metadata only; excluded from lead score	Not scored

5. COASTAL HAZARD & BEACHING INTERCEPTION ALERT

MODELLED SHORELINE ALERT: Hydrodynamic forecasting estimates possible slick impact along the coastline within 11.2 hours. Interception coordinates: 22.5800° N, 69.4500° E. This scenario output should be reviewed by the responsible incident commander before any response decision.

5. ANALYST NEXT STEPS

- 1. Obtain original calibrated SAR/EO products and preserve their source metadata and hashes.
- 2. Re-run drift with authoritative, time-aligned current and wind fields and report uncertainty bounds.
- 3. Verify AIS completeness, vessel identity, and timing against the original provider export; obtain independent corroboration before contacting vessel **MT NEPTUNE GLORY (IMO: 9384722)**.
- 4. Refer any enforcement decision to the competent authority and applicable law.

REPRODUCIBILITY FINGERPRINT — NOT A LEGAL CERTIFICATE:
This SHA-256 value fingerprints the displayed screening summary at generation time. It does not authenticate source imagery, establish chain of custody, or substitute for a qualified certificate or authority review.
Summary manifest SHA-256: e9dcd72e72098ba1f8478fce27c323b1e61e3124ccfd31513df7ecfdfe95267e
No source provenance was supplied.